MACHINE LEARNING FOR SIGNAL PROCESSING

5-2-2025

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STORY SO FAR

Classification Problem

Decision Theory

Generative Modeling

Function Modeling

Discriminative Modeling

Gaussian Modeling

Gaussian Mixture Modeling

EM algorithm

Linear Models for Regression and Classification

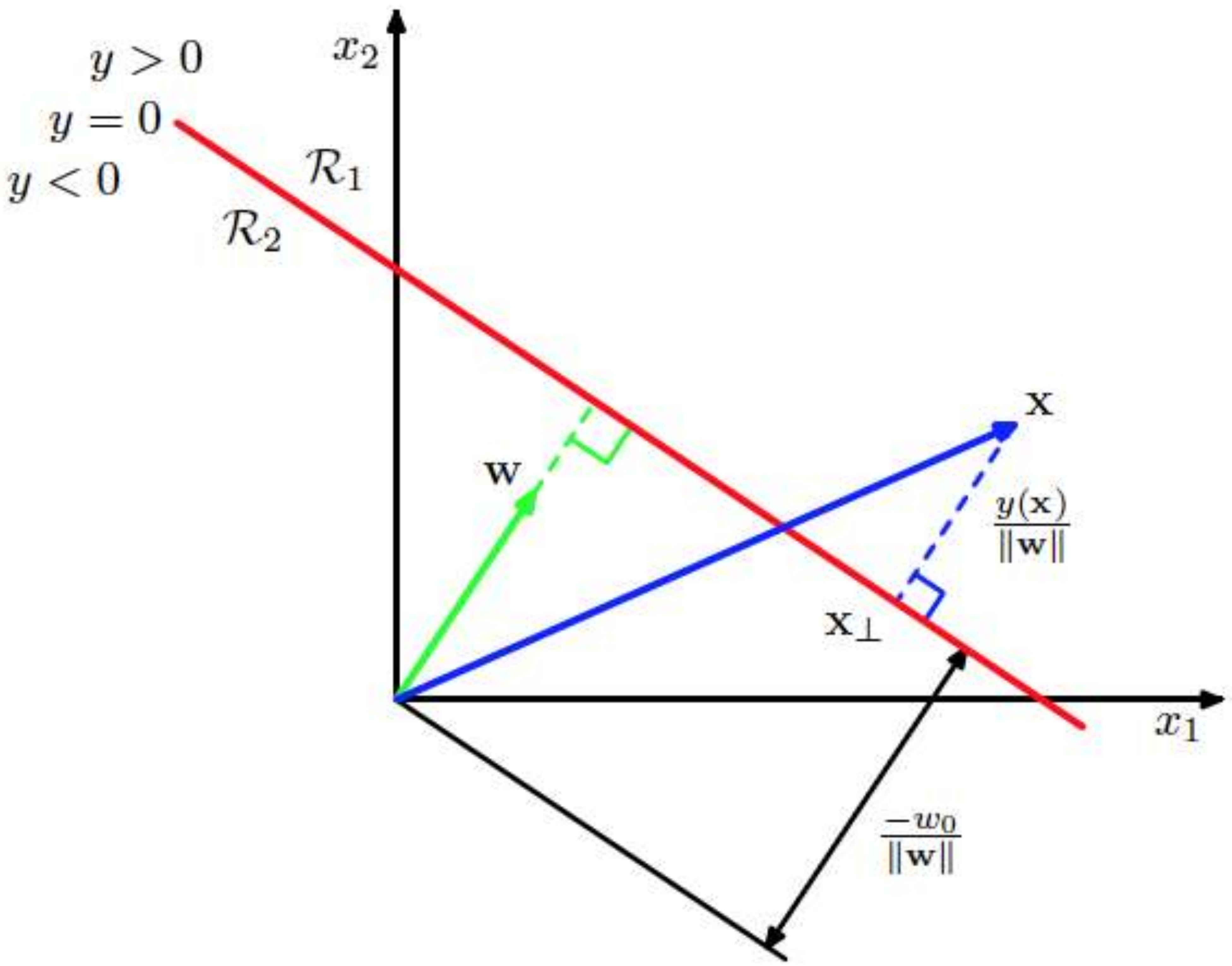
Gradient Descent Algorithm

Data Representations PCA, LDA

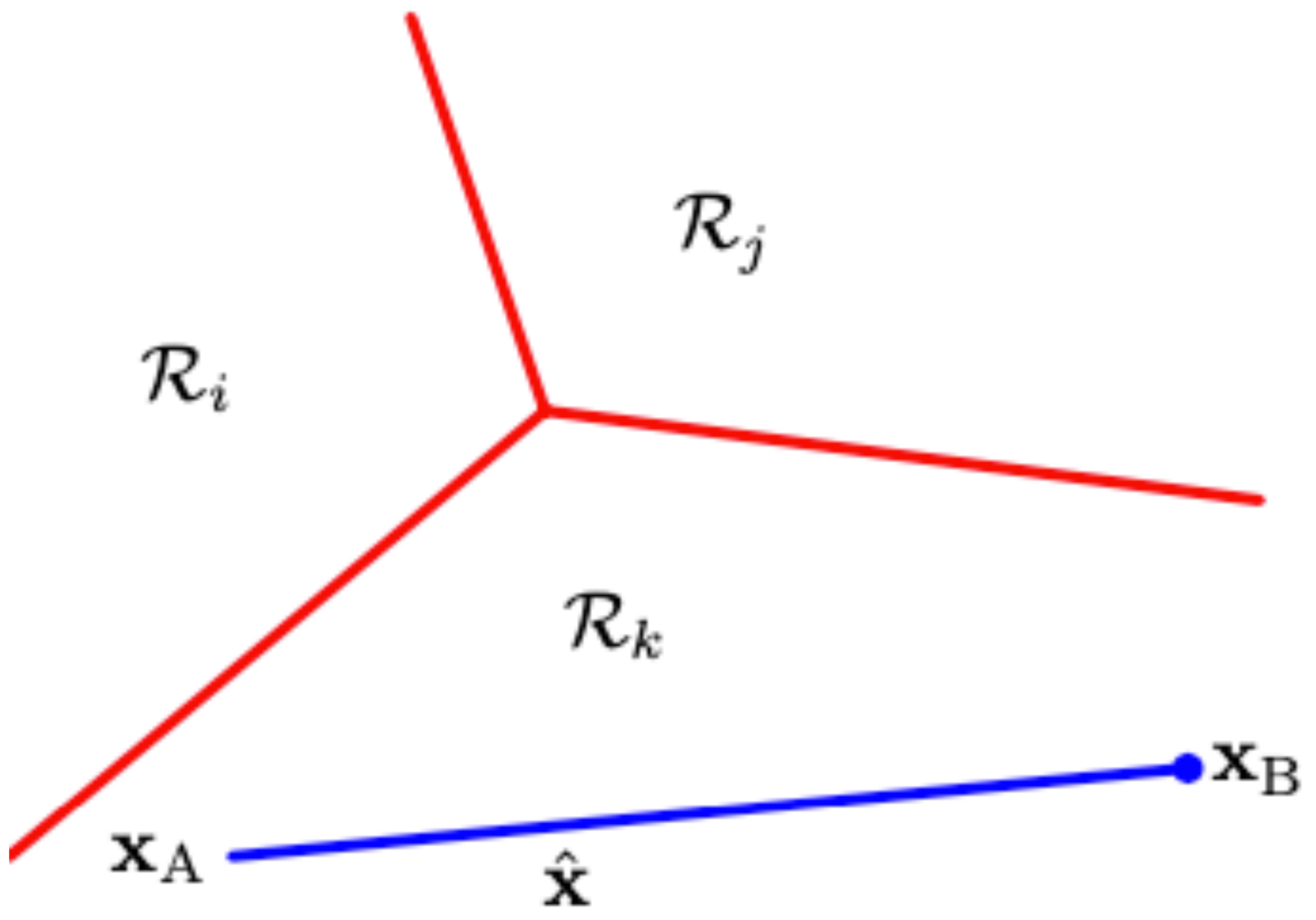
Linear Models for Classification

- Optimize a modified cost function

$$y(\mathbf{x}) = \mathbf{w}^{\mathrm{T}}\mathbf{x} + w_0$$



LINEAR MODELS FOR CLASSIFICATION



Least Squares for Classification

- K-class classification problem

$$y_k(\mathbf{x}) = \mathbf{w}_k^{\mathrm{T}}\mathbf{x} + w_{k0}$$

$$\mathbf{y}(\mathbf{x}) = \widetilde{\mathbf{W}}^{\mathrm{T}}\widetilde{\mathbf{x}}$$

- With 1-of-K hot encoding, and least squares regression

$$E_D(\widetilde{\mathbf{W}}) = \frac{1}{2}\mathrm{Tr}\left\{(\widetilde{\mathbf{X}}\widetilde{\mathbf{W}} - \mathbf{T})^{\mathrm{T}}(\widetilde{\mathbf{X}}\widetilde{\mathbf{W}} - \mathbf{T})\right\}$$

Logistic Regression

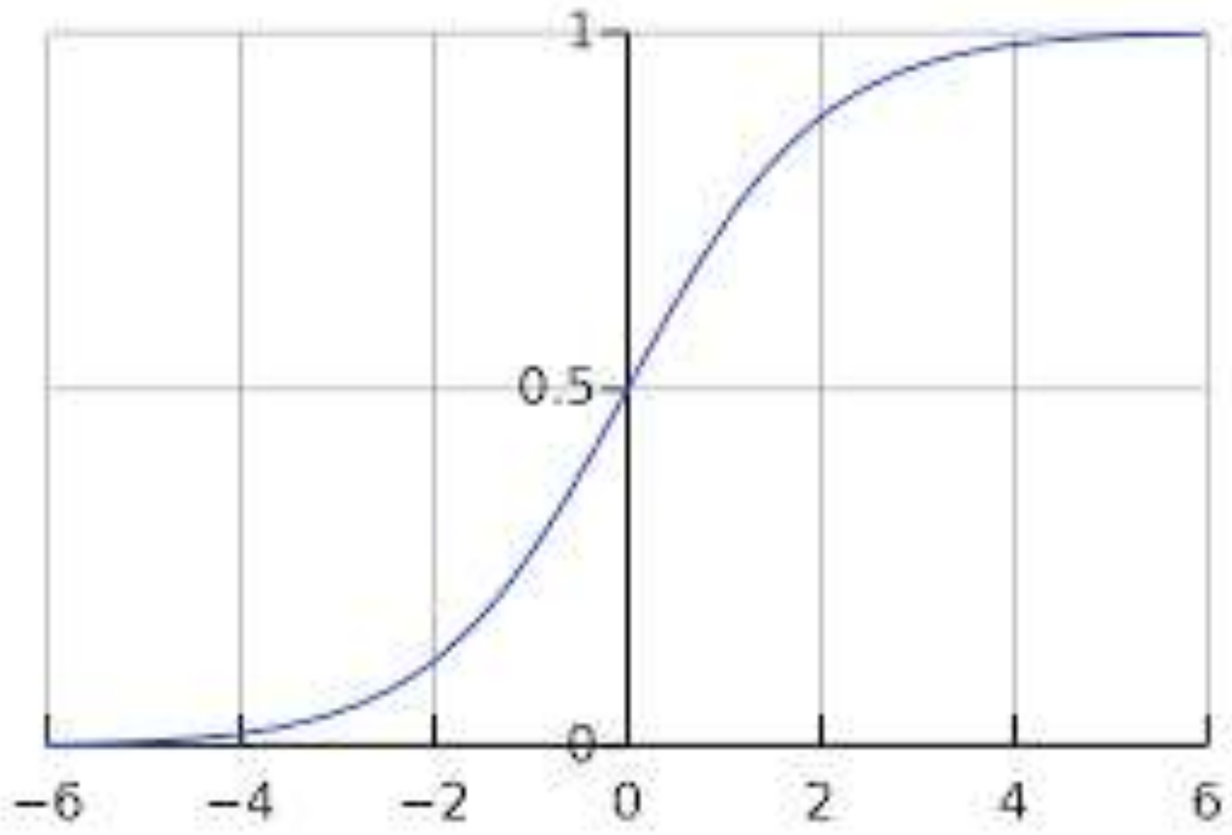
$$p(\mathcal{C}_1|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1) + p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} = \frac{1}{1+\exp(-a)} = \sigma(a)$$

where we have defined

$$a = \ln \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)}$$

and $\sigma(a)$ is the *logistic sigmoid* function defined by

$$\sigma(a) = \frac{1}{1+\exp(-a)}$$



Logistic Regression

- 2- class logistic regression

$$p(\mathcal{C}_1|\phi) = y(\phi) = \sigma\left(\mathbf{w}^{\mathrm{T}}\phi\right)$$

  - Maximum likelihood solution $$\nabla E(\mathbf{w}) = \sum_{n=1}^{N}(y_n - t_n)\phi_n$$

  - K-class logistic regression $$p(\mathcal{C}_k|\phi) = y_k(\phi) = \frac{\exp(a_k)}{\sum_j \exp(a_j)}$$

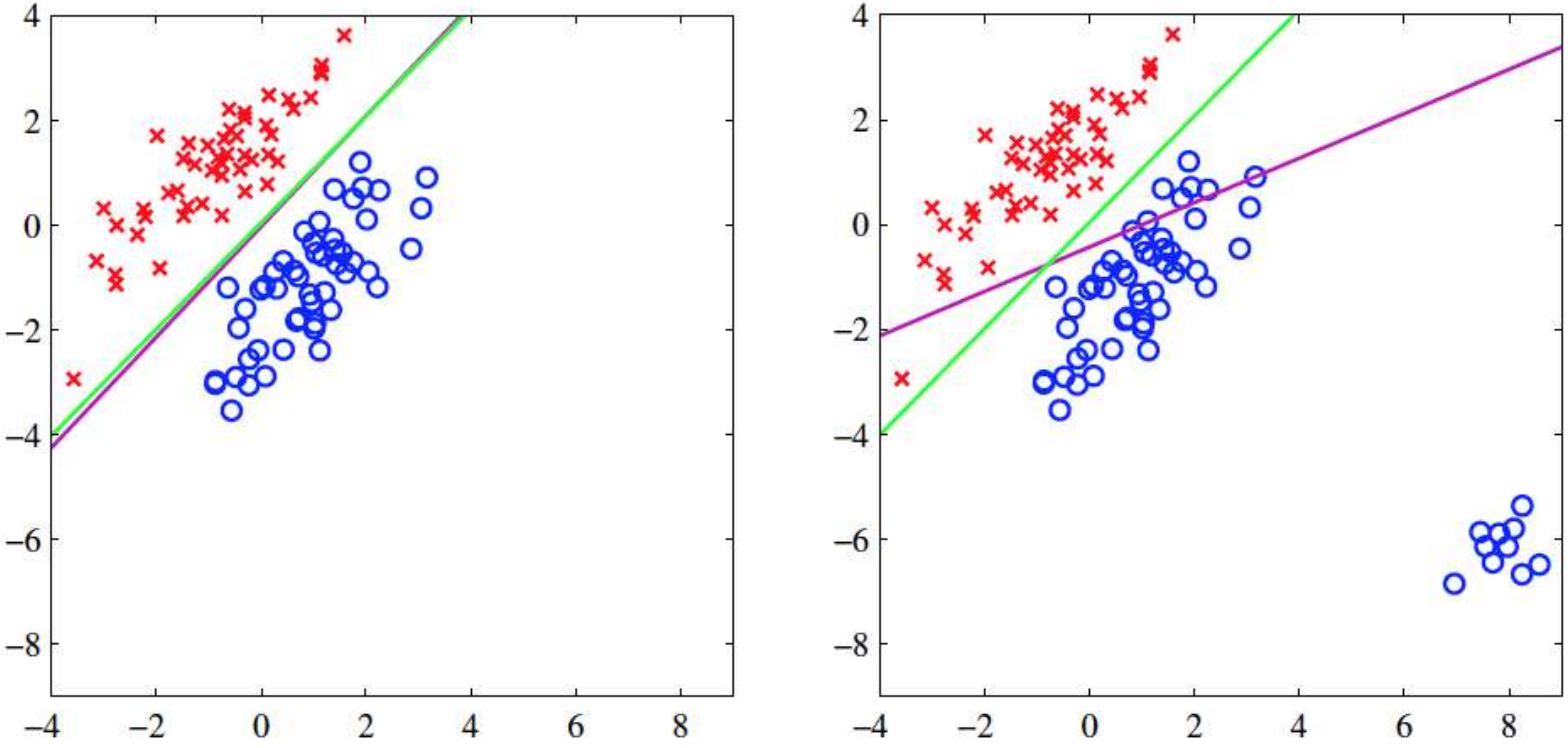
$$a_k = \mathbf{w}_k^{\mathrm{T}}\phi.$$

  - Maximum likelihood solution

$$\nabla_{\mathbf{w}_j} E(\mathbf{w}_1, \ldots, \mathbf{w}_K) = \sum_{n=1}^{N}(y_{nj} - t_{nj})\,\phi_n$$

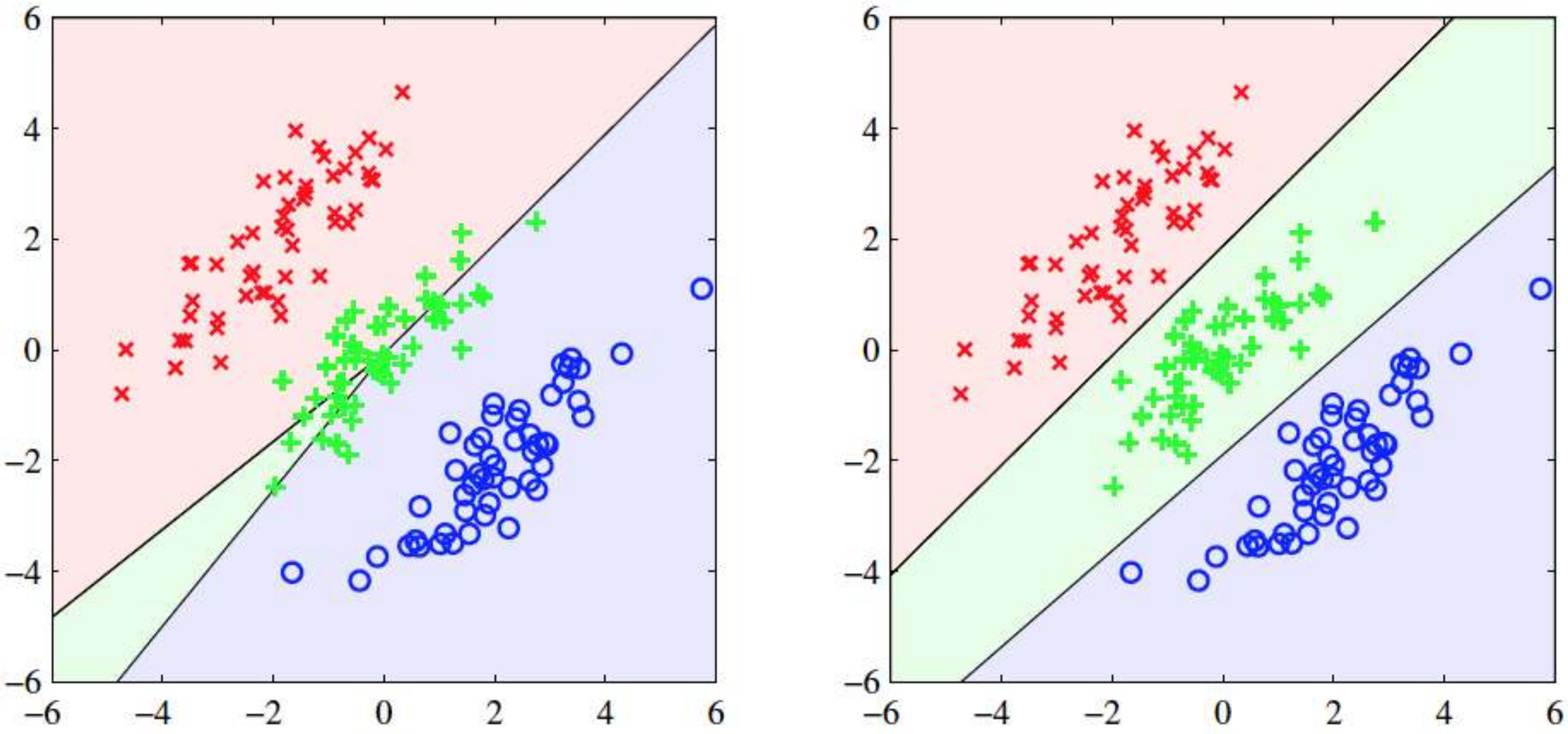
Least Squares versus Logistic Regression

❖ 2- class logistic regression



Least Squares versus Logistic Regression

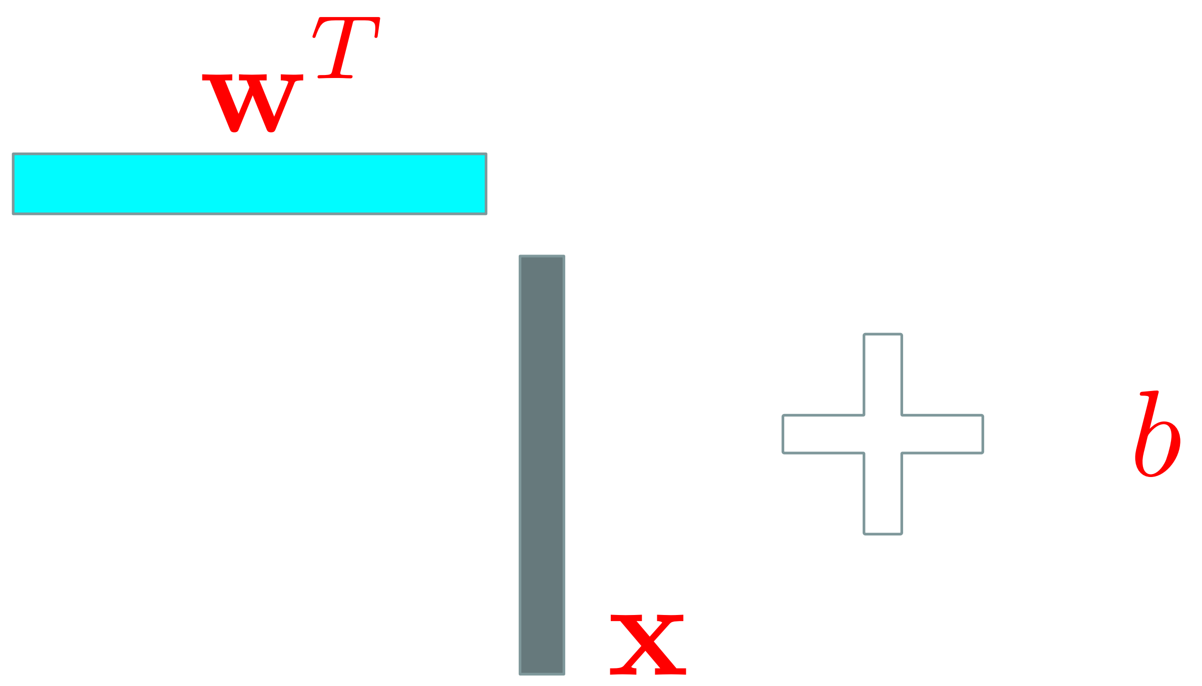
- 2- class logistic regression



Lets Start Off With Simplest Model

- Define the model as an affine transform

$$\mathbf{w}^T\mathbf{x} + b$$



$$w_1x_1 + w_2x_2 + w_3x_3 + ...w_Dx_D + b$$

- Convert the values to probabilities

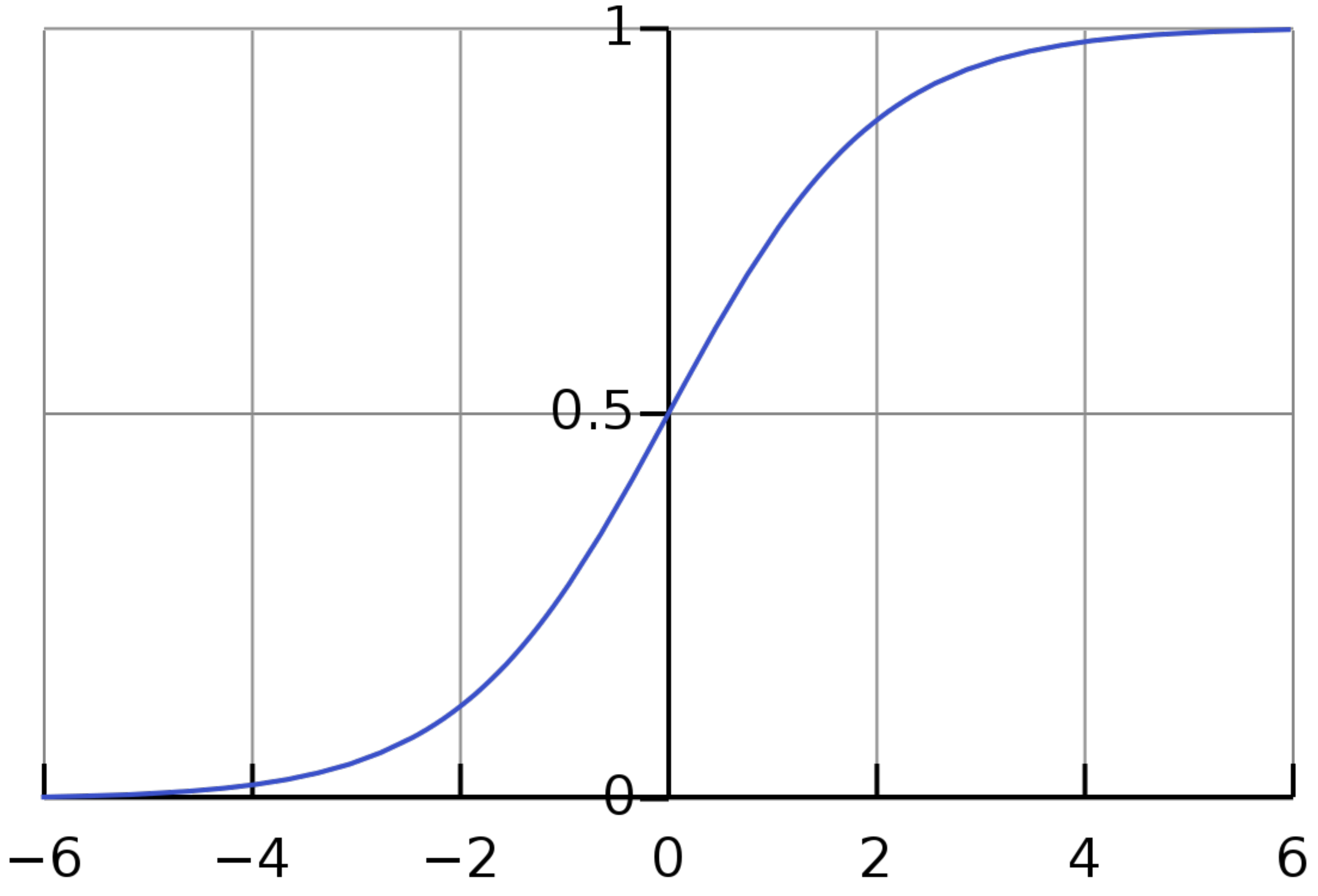
$$\mathbf{w}^T\mathbf{x} + b \geq 0 \rightarrow x \in C_1 \qquad p(C_1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b)$$

$$\mathbf{w}^T\mathbf{x} + b < 0 \rightarrow x \in C_2 \qquad p(C_2|\mathbf{x}) = 1 - p(C_1|\mathbf{x})$$

LETS START OFF WITH SIMPLEST MODEL

- Sigmoid function

$$\sigma(a) = \frac{1}{1+e^{-a}}$$



- Convert the values to probabilities

$$\mathbf{w}^T\mathbf{x} + b \geq 0 \rightarrow x \in C_1 \qquad p(C_1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b)$$

$$\mathbf{w}^T\mathbf{x} + b < 0 \rightarrow x \in C_2 \qquad p(C_2|\mathbf{x}) = 1 - p(C_1|\mathbf{x})$$

How do we find the model parameters

- Given a set of training data

  - The targets

$$\{\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_N\}$$

$$\{t_1, t_2, \ldots t_N\}$$

$$t_n = 1 \quad for \quad \mathbf{x}_n \in \mathbf{C}_1$$

$$t_n = 0 \quad for \quad \mathbf{x}_n \in \mathbf{C}_2$$

  - The model outputs

$$y_n = \sigma(\mathbf{w}^T \mathbf{x}_n + b)$$

  - probability when

$$t_n = 1 \rightarrow y_n^{t_n}$$

  - probability when

$$t_n = 0 \rightarrow (1 - y_n)^{(1-t_n)}$$

  - Jointly

$$p(y_n) = y_n^{t_n}(1 - y_n)^{(1-t_n)}$$

TOTAL PROBABILITY

- Each data sample is independent

$$p([y_1, y_2 ..., y_N]) = p(y_1)p(y_2)...p(y_N)$$

$$\begin{aligned} p([y_1, y_2 ..., y_N]) &= p(y_1)p(y_2)...p(y_N) \\ &= \prod_{n=1}^{N} y_n^{t_n}(1 - y_n)^{(1-t_n)} \end{aligned}$$

$$y_n = \sigma(\mathbf{w}^T \mathbf{x}_n + b)$$

- Log total probability $$log\ p([y_1, y_2 ..., y_N]) = \sum_{n=1}^{N} t_n log\ y_n + (1 - t_n) log\ (1 - y_n)$$

HOW DO WE FIND THE MODEL PARAMETERS

- We want to maximize the total probability
  - Or equivalently minimize the negative log probability

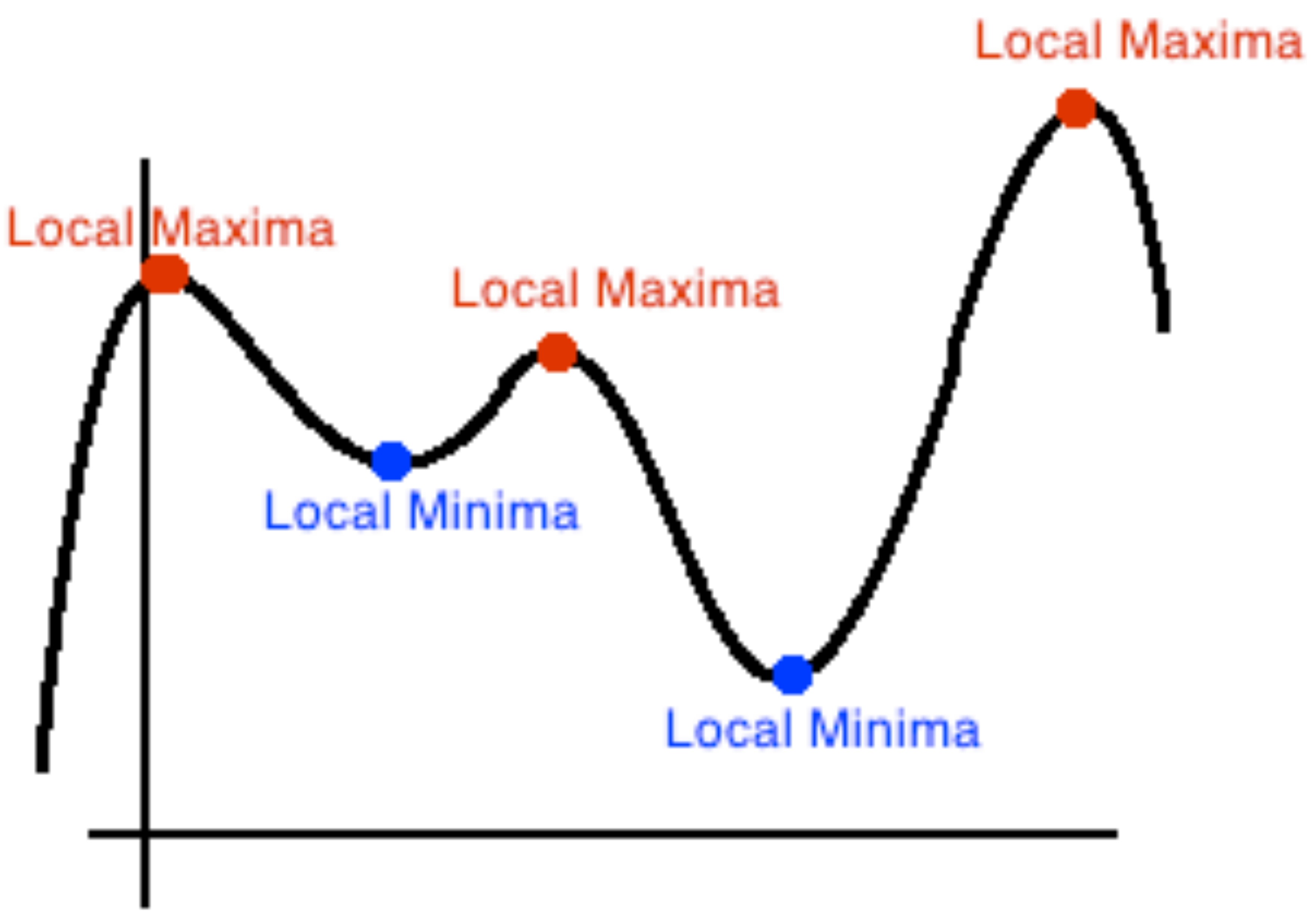
$$E(\mathbf{w}, b) = -\left[\sum_{n=1}^{N} t_n log\ y_n + (1 - t_n) log\ (1 - y_n)\right]$$

$$y_n = \sigma(\mathbf{w}^T \mathbf{x}_n + b)$$

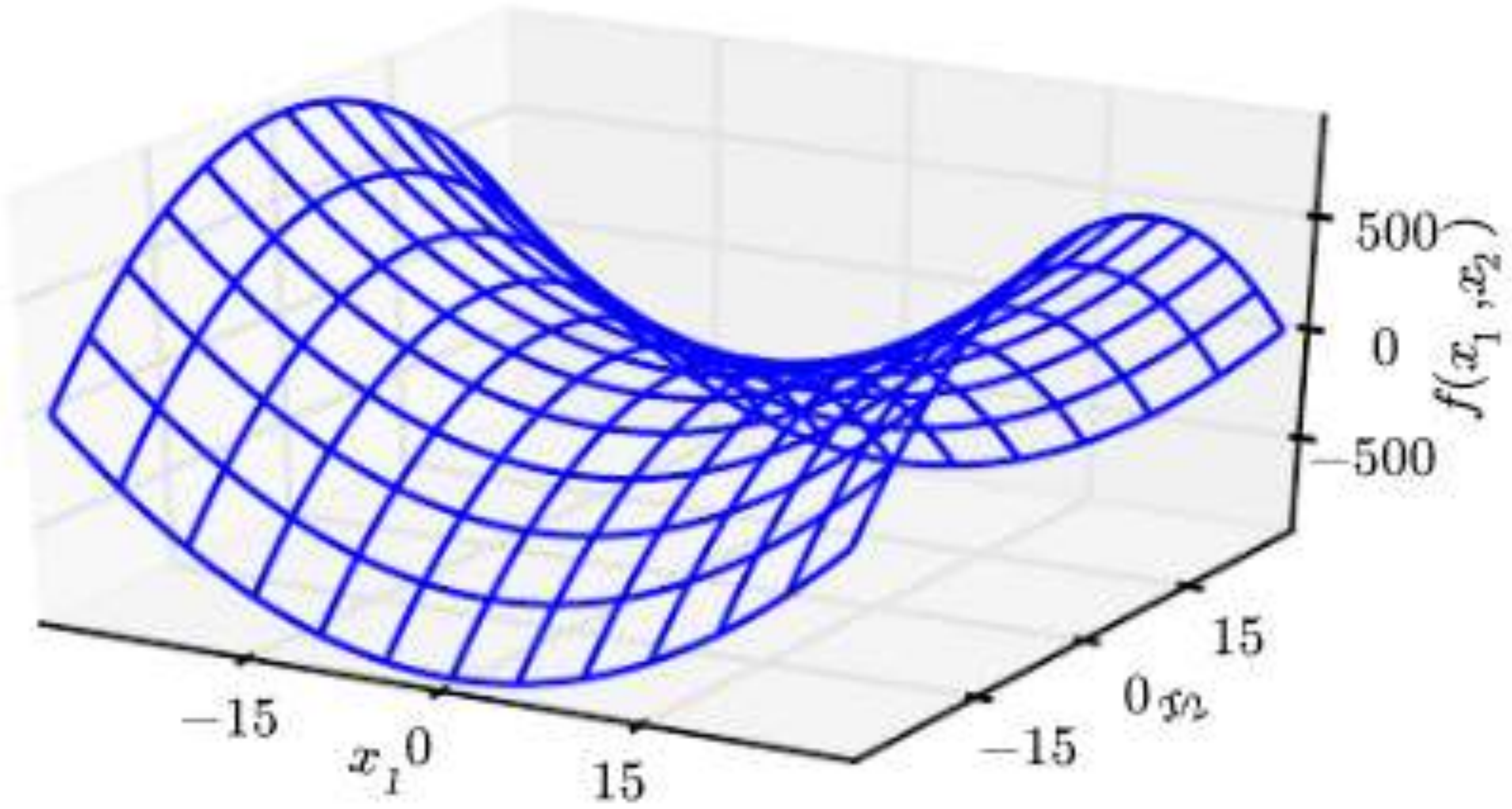
- The parameters of the model need to solved based on the minimizing the error
  - This error function is non convex in the parameters
    - Need to perform iterative update —> Gradient descent

NON-CONVEX OPTIMIZATION

- Local maxima and minima

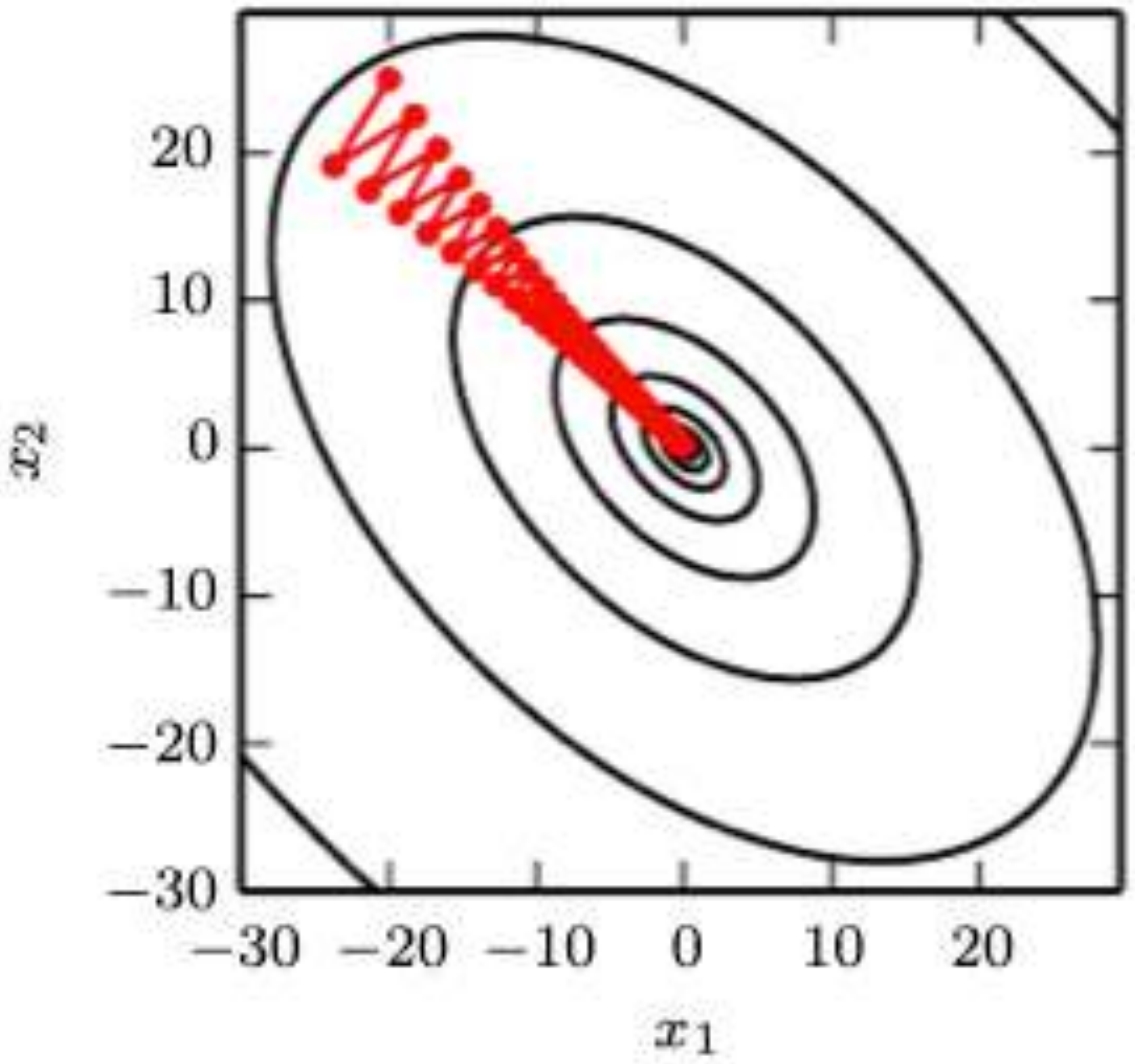


Gradient Descent



Multi-dimensional saddle points

GRADIENT DESCENT



Slowness of gradient descent

For a quadratic function

Gradient Descent Algorithm

- Start with random weights

$$iter \quad t = 0 \rightarrow \mathbf{w}^0, b^0$$

- Compute gradients and update the model iteratively

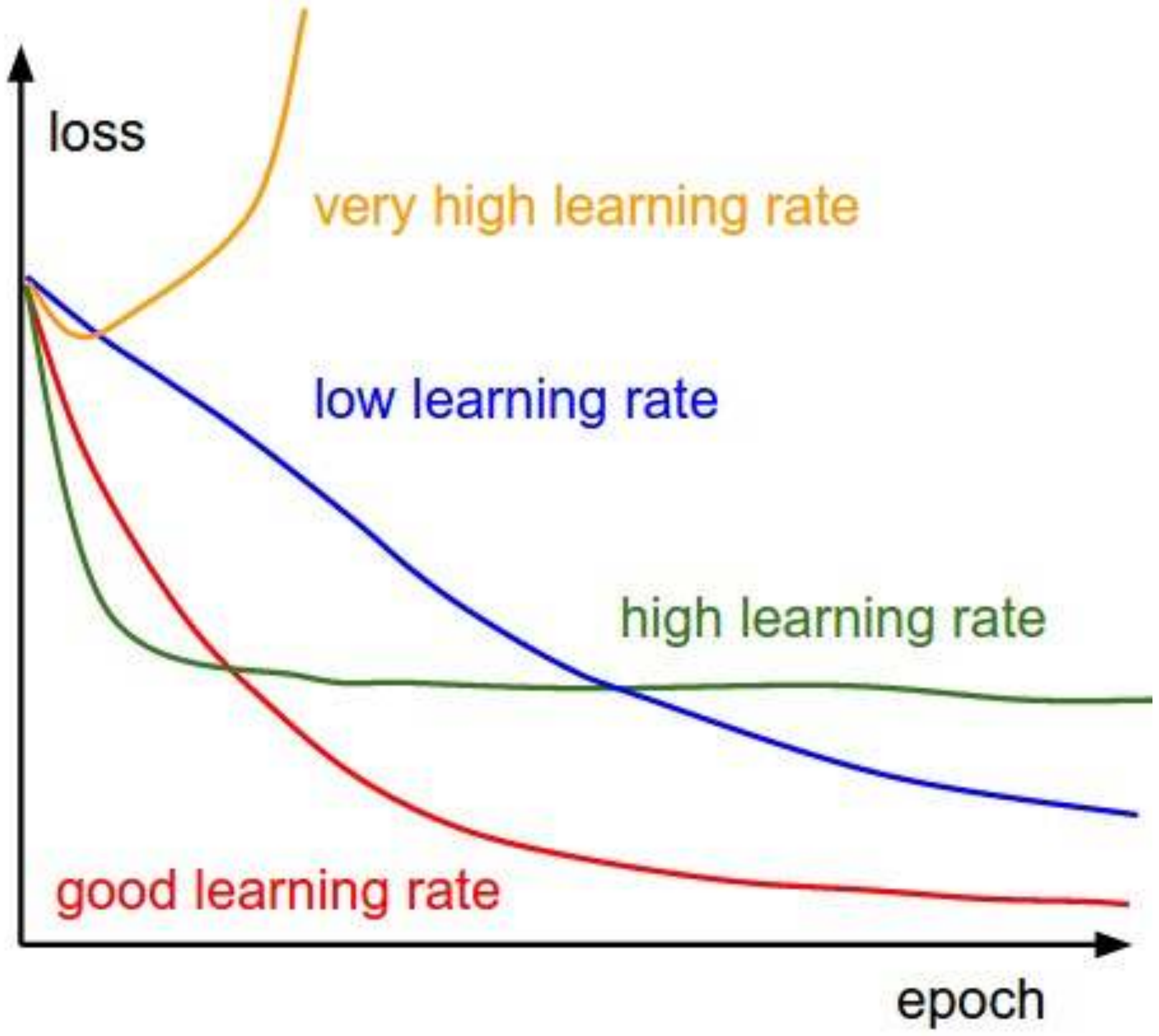
$$\mathbf{w}^t = \mathbf{w}^{t-1} - \eta \frac{\partial E}{\partial \mathbf{w}}\bigg|_{\mathbf{w},b=\mathbf{w}^{t-1},b^{t-1}}$$

$$b^t = b^{t-1} - \eta \frac{\partial E}{\partial b}\bigg|_{\mathbf{w},b=\mathbf{w}^{t-1},b^{t-1}}$$

- Check for convergence —> If not, $t = t + 1$ and go to the step above.

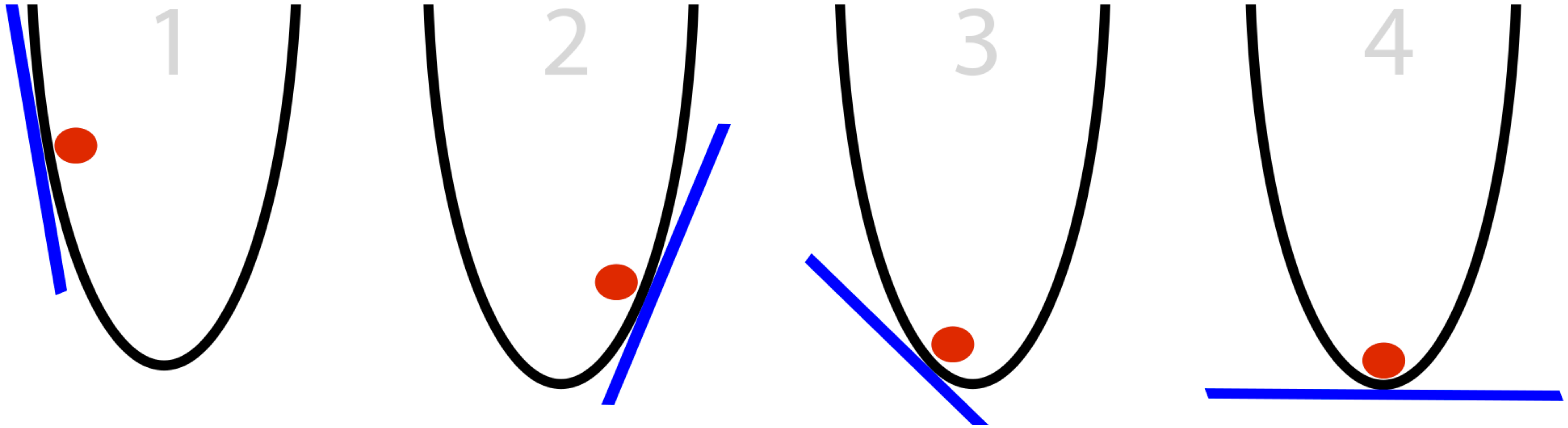
LEARNING RATE

- Solving a non-convex optimization.
- Iterative solution.
- Depends on the initialization.
- Convergence to a local optima.



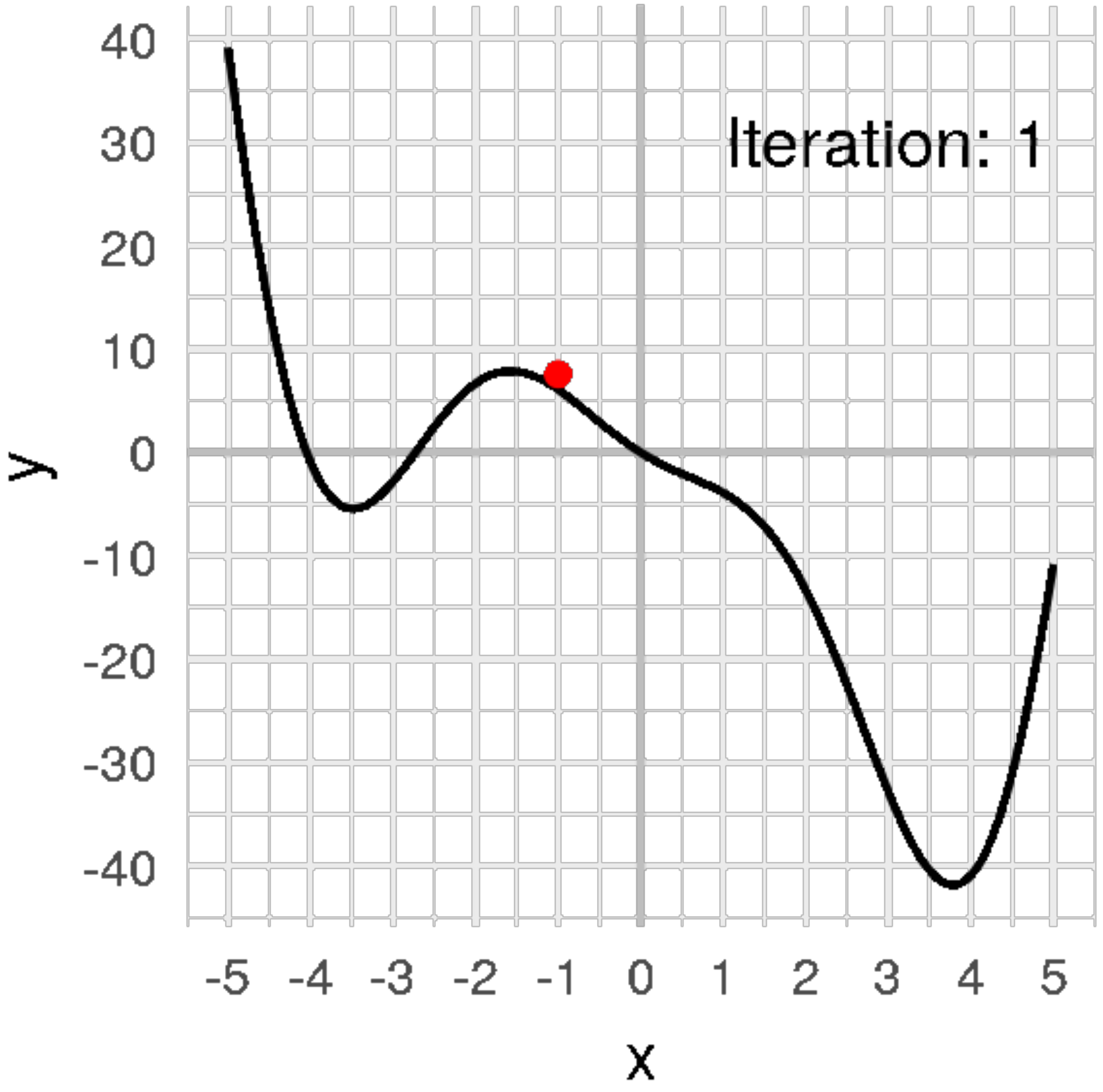
ERROR FUNCTION REVISITED

- Iterative solution is also possible.
  - ✓ Move the parameters in the negative direction of the gradient

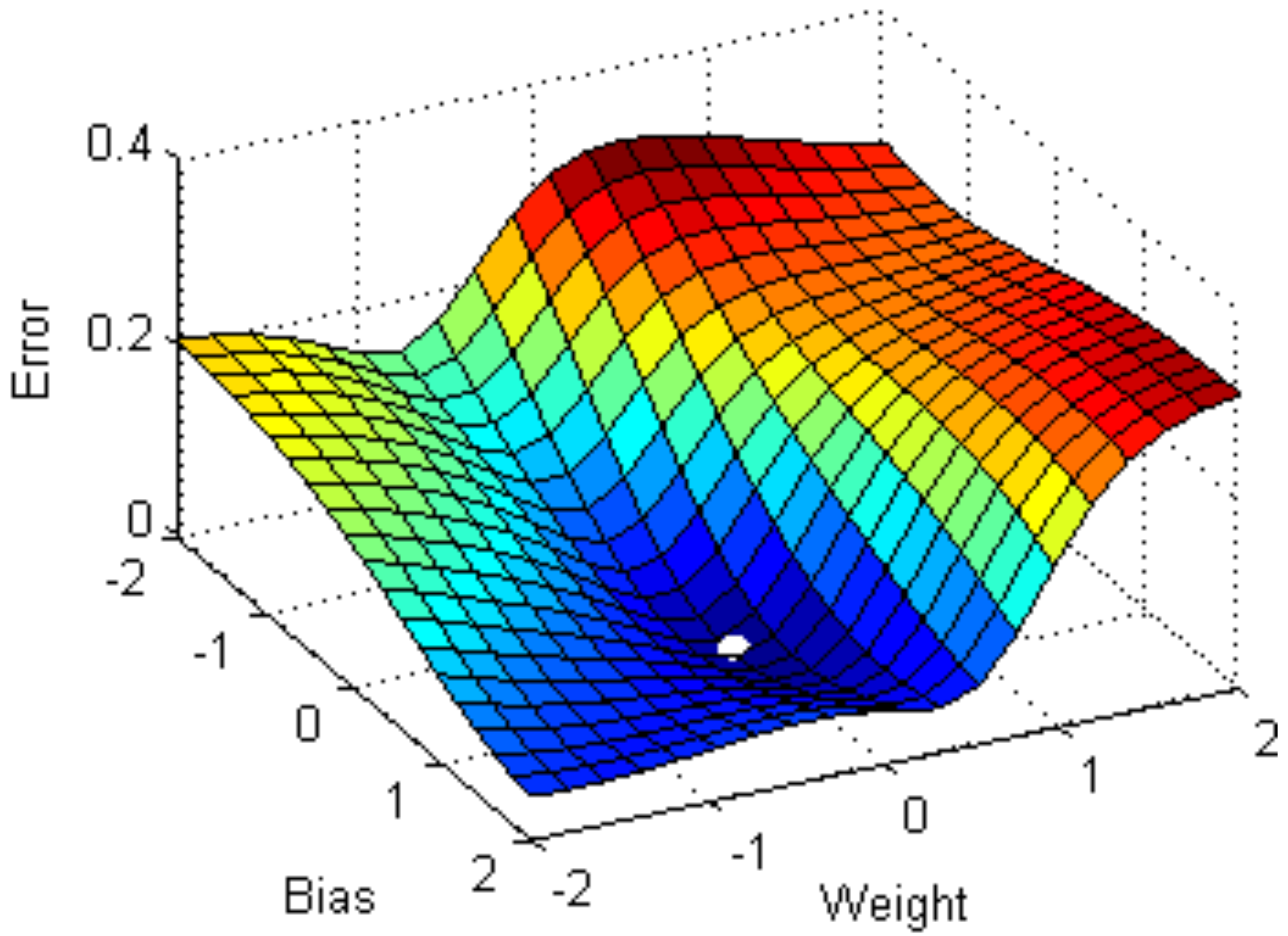


Negative gradient Positive gradient Negative gradient Zero gradient

Gradient Descent Algorithm



Gradient Descent Algorithm



BACK TO OUR PROBLEM

MULTI-CLASS LOGISTIC REGRESSION

- Outputs are

$$\mathbf{y} = softmax(\mathbf{W}^T\mathbf{x} + \mathbf{b})$$

- Targets are one hot encoded vectors

$$\mathbf{t} \quad \begin{bmatrix} 0 \\ 0 \\ . \\ 0 \\ 1 \\ 0 \\ . \\ 0 \end{bmatrix}$$

- Error function

$$E(\mathbf{W}, \mathbf{b}) = -\sum_n \sum_k t_{nk} \; log \; y_{nk}$$

  - cross entropy function
  - as before, learnt using gradient descent

STOCHASTIC GRADIENT DESCENT

- The total error is a sum of errors in all the data samples

$$E(\mathbf{w}, b) = -\left[\sum_{n=1}^{N} t_n log\ y_n + (1 - t_n) log\ (1 - y_n)\right]$$

  - May be too slow - requiring a large number of updates before convergence.

- Can we take a subset of the data and perform gradient descent and update the model on the subset ?

Stochastic Gradient Descent

- Split the data into random chunks

- perform gradient updates on each mini batch
  - assumption - gradients computed on the mini-batch well approximate the full batch gradients

STOCHASTIC GRADIENT DESCENT

- Split the data into random chunks

- hyper-parameters
  - batch size

Linear Regression Revisited

- Primal and dual forms
- Solution in dual space
- Kernels

THANK YOU

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